Parth Sachin Deshmukh

parthdeshmukh321@gmail.com | (979) 492-9902 | LinkedIn | GitHub | HuggingFace | Portfolio

Education

Texas A&M University — M.S. Computer Science — GPA: **4.0/4.0** May 2027 (expected)
Coursework: Distributed Systems (in progress), Artificial Intelligence, Natural Language Processing, Analysis of Algorithms

Vishwakarma Institute of Information Technology (VIIT) — B.Tech. IT — GPA: **9.12/10.00** May 2024

Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Java

AI & LLMs: OpenAI SDK, Claude, LangChain, LangGraph, MCP, LLM tool calling, prompt engineering, generative AI, NLP, RAG (BM25 + dense), embeddings, vector databases (Qdrant, Pinecone), LLM evaluation (LLM-as-judge), Azure OpenAI

ML & Data: PyTorch, scikit-learn, TensorFlow, statistical analysis, data visualization

Backend & Data: FastAPI, REST APIs, Flask, Pydantic (data modeling), React, PostgreSQL, SQLite, Kafka, Temporal, Neo4j

Cloud & Tools: Docker, AWS (EC2, ECR), GitHub Actions (CI/CD), Git, Linux, unit testing (pytest), debugging, OpenTelemetry, Langfuse, Cursor

Experience

ML/Data Research Intern — Indian Institute of Technology (IIT) Bombay Feb 2024 – Sep 2024

- Gathered requirements from **20+** faculty and built the analytics web app they used for reporting (JavaScript, AJAX, PHP, MySQL).
- Indexed MySQL queries across **500+** report configs and added async loading, cutting report latency **~40–80%**.
- Wrote a Figma-to-HTML parser behind **100+** reusable reporting themes; trained faculty and turned their feedback into fixes.

Projects

Dream Lab Student Simulator *GitHub* Jun 2026 – Present

- Compared **5** student LLMs on **32** scenarios with rule checks and LLM judges: all **100%** on rules, consistency **53.6–75.0%**.
- Fine-tuned Llama 3.1 8B with 4-bit QLoRA (PyTorch, PEFT, TRL) on **768** dialogues in **~4** h on TAMU HPRC, reaching **74.5%** LLM-judged consistency.
- Replaced a regex parser with an LLM parser + regex veto, raising exact match to **91.7%** from **81.7%** on **60** hand-labeled cases.
- Wrote **29** pytest modules (**312** tests) that catch behavior regressions as prompts and models change.

RAG-Powered Code Assistant *GitHub* Mar 2026

- Replaced LangChain's prebuilt agent with a **141**-line tool-calling loop capped at **6** steps, **8** tool calls and **10** s timeouts.
- Ran a retrieval eval on **220** auto-generated questions: hybrid BM25 + dense hit **0.88** recall@5 vs **0.55**; rejected a reranker (**0.82**).
- Shipped **106** tests at **93%** coverage and a GitHub Actions workflow that deploys the Docker app to Hugging Face Spaces.

Realtime Multi-Agent Knowledge Graph *GitHub* Jun 2026 – Present

- Ran each Kafka speech-to-text segment as a Temporal workflow through **4** LangGraph agents, with at most **5** running at once.
- Gated Neo4j writes behind an MCP commit tool (approved entities only, MERGE upserts) with **240** s activity timeouts.
- Served Neo4j Cypher + Qdrant retrieval through FastAPI and traced every stage with OpenTelemetry + Langfuse.

Career Ops — Job Ingest & Resume Tooling Jun 2026 – Present

- Built with Cursor and Claude under AGENTS.md rules; every AI-written diff must pass **800+** pytest tests and ATS parse gates.
- Integrated **10+** job sources with bounded concurrency and 429/5xx backoff; deduped postings with Pydantic models and SQLite.

Publications

- **First Author**, Springer 2024: *Automated GUI Testing to Enhance UX: A Survey of the State of the Art*
- Co-Author, IEEE 2024: *Increasing Donation Transparency in Disaster Relief: A Blockchain-based Solution*